Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The solution of $\frac{dy}{dx} = |x|$ is :

(A) $y = \frac{x|x|}{2} + c$

(B) $y = \frac{|x|}{2} + c$

(C) $y = \frac{x^2}{2} + c$

(D) $y = \frac{x^3}{2} + c$

Where c is an arbitary constant

Q.2

The solutions of $(x + y + 1)dy = dx$ are

(A) $x + y + 2 = Ce^y$

(B) $x + y + 4 = C \log y$

(C) $\log(x + y + 2) = Cy$

(D) $\log(x + y + 2) = C - y$

Q.3

The curve satisfying the equation $\frac{dy}{dx} = \frac{y\left(x+y^3\right)}{x\left(y^3-x\right)}$ and passing through the point $(4, -2)$ is

(A) $y^2 = -2x$

(B) $y = -2x$

(C) $y^3 = -2x$

(D) None of these

Q.4

The particular solution of the differential equation $\sin^{-1}\left(\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2} - 1\right) = \mathrm{x}$, where $y = \frac{dy}{dx} = 0$ when $x = 0$, is

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(A) $y = x^2 + x - \sin x$

(B) $y = \frac{x^2}{2} + x - \sin x$

(C) $y = \frac{x^2}{2} + \frac{x}{2} - \sin x$

(D) $2y = x^2 + x - \sin x$

Q.5

The solution of $(y + x + 5)dy = (y - x + 1)dx$ is

(A) $\log\left((y+3)^2 + (x+2)^2\right) + \tan^{-1}\frac{y+3}{y+2} + C$

(B) $\log\left((y+3)^2 + (x-2)^2\right) + \tan^{-1}\frac{y-3}{x-2} = C$

(C) $\log\left((y+3)^2 + (x+2)^2\right) + 2\tan^{-1}\frac{y+3}{x+2} = C$

(D) $\log\left((y+3)^2 + (x+2)^2\right) - 2\tan^{-1}\frac{y+3}{x+2} = C$

Q.6

The equation of the curve satisfying
$xdy - ydx = \sqrt{x^2 - y^2}$ and $y(1) = 0$ is :

(A) $y = x^2 \log(\sin x)$

(B) $y = x \sin(\log x)$

(C) $y^2 = x(x-1)^2$

(D) $y = 2x^2(x-1)$

Q.7

$y + x\frac{dy}{dx} = x\frac{\phi(xy)}{\phi'(xy)}$ then $\phi(xy)$ is equation to

(A) $ke^{x^2/2}$

(B) $ke^{y^2/2}$

(C) $ke^{xy/2}$

(D) ke^{xy}

Q.8

If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is

(A) $\log\left(\frac{y}{x}\right) = cx$

(B) $\log\left(\frac{x}{y}\right) = cy$

(C) $y\log\left(\frac{x}{y}\right) = cy$

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(D) $x \log\left(\frac{y}{x}\right) = cy$

Q.9

The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi(y/x)}{\phi'(y/x)}$ is

(A) $x\phi(y/x) = k$

(B) $\phi(y/x) = kx$

(C) $y\phi(y/x) = k$

(D) $\phi(y/x) = ky$

(k is arbitrary constant)

Q.10

The real value of n for which the substitution $y = u^n$ will
transform the differential equation $2x^4y\frac{dy}{dx} + y^4 = 4x^6$ into a homogeneous equation is

(A) $1/2$

(B) 1

(C) $3/2$

(D) 2

Q.11

The solution of the equation $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$ is

(A) $\sin^{-1} y - \sin^{-1} x = c$

(B) $\sin^{-1} y \sin^{-1} x = c$

(C) $\sin^{-1}(xy) = 2$

(D) None of these

Q.12

A continuously differentiable function $\phi(x)$,
$x \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$ satisfying
$y' = 1 + y^2, y(0) = 0 = y(\pi)$ is

(A) $\tan x$

(B) $x(x - \pi)$

(C) $(x - \pi)(1 - e^x)$

(D) $\sec^2 x$

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Q.13

The solution of the equation
$x\int_0^x y(t)dt = (x+1)\int_0^x ty(t)dt, x > 0$ is

(A) $\mathrm{y} = \frac{\mathrm{c}}{\mathrm{x}^3}\mathrm{e}^{\mathrm{x}^3}$

(B) $y = cx^3e^{-x^3}$

(C) $\frac{\mathrm{c}}{\mathrm{x}^3}\mathrm{e}^{-\frac{1}{\mathrm{x}}}$

(D) None of these

Q.14

The general solution of the differential equation
$\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2} = \cos \mathrm{nx}$ is

(A) $n^2y + \cos nx = n^2(Cx + D)$

(B) $n^2y - \sin nx = n^2(-Cx + D)$

(C) $n^2y + \cos nx = \frac{Cx+D}{n^2}$

(D) None of these. [Where C and D are arbitrary constants]

Q.15

What is the solution of the differential equation
$\sin\left(\frac{dy}{dx}\right) - a = 0?$

(A) $y = x\sin^{-1}a + c$

(B) $x = y\sin^{-1}a + c$

(C) $y = x + x\sin^{-1}a + c$

(D) $y = \sin^{-1}a + c$
where c is an arbitrary constant.

Q.16

What is the solution of the equation
$\ln\left(\frac{dy}{dx}\right) + x = 0?$

(A) $y + e^x = c$

(B) $y - e^{-x} = c$

(C) $y + e^{-x} = c$

(D) $y - e^x = c$

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Q.17

$y = y(x)$ and $\frac{2+\sin x}{1+y}\left(\frac{dy}{dx}\right) = -\cos x, y(0) = 1$
then $y\left(\frac{\pi}{2}\right)$ equals

(A) $1/3$

(B) $2/3$

(C) $-1/3$

(D) 1

Q.18

The solution of the differential equation $\mathrm{d}x\left(x^2 - \mathrm{t}\right) = \mathrm{dt}$, is

(A) $\mathrm{t} = (x-1)^2\mathrm{e}^x + \mathrm{c}$

(B) $\mathrm{t} = (x+1)^2\mathrm{e}^x + \mathrm{c}$

(C) $\mathrm{t} = \left(x^2 + 2x - 1\right)\mathrm{e}^x + \mathrm{c}$

(D) None of these

Q.19

The solution of the equation $\frac{dy}{dx} = \frac{x(2\log x+1)}{\sin y+y\cos y}$ is

(A) $y\sin y = x^2\log x + \frac{x^2}{2} + c$

(B) $y\cos y = x^2(\log x + 1) + c$

(C) $y\cos y = x^2\log x + \frac{x^2}{2} + c$

(D) $y\sin y = x^2\log x + c$

Q.20

The solution of $\left(\frac{xdx+ydy}{xdy-ydx}\right) = \sqrt{\left(\frac{a^2-x^2-y^2}{x^2+y^2}\right)}$ is

(A) $\sqrt{(x^2+y^2)} = a\sin\left\{\left(\tan^{-1}y/x\right) + \text{constant}\right\}$

(B) $\sqrt{(x^2+y^2)} = a\cos\left\{\left(\tan^{-1}y/x\right) + \text{constant}\right\}$

(C) $\sqrt{(x^2+y^2)} = a\left\{\tan\left(\sin^{-1}y/x\right) + \text{constant}\right\}$

(D) $\sqrt{(x^2+y^2)} = a\left\{\tan\left(\cos^{-1}y/x\right) + \text{constant}\right\}$

Answers & Solutions

Q.1 Answer:

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$y = \frac{x|x|}{2} + c$

Solution:

$\frac{dy}{dx} = |x|$
$\frac{dy}{dx} = x$ **for** $x \geq 0$; $\frac{dy}{dx} = -x$ **for** $x < 0$; $\int dy = \int x dx$
$y = \frac{x^2}{2} + C_1$
$\int \mathrm{dy} = -1\mathrm{xdx}$
$y = -\frac{x^2}{2} + C_1$
From (i) and (ii) $y = \frac{x|x|}{2} + C$

Q.2 Answer:

$x + y + 2 = Ce^y$

Solution:

Putting $x + y + 1 = u$**, we have** $du = dx + dy$ **and the given equation reduces to** $u(du - dx) = dx \Rightarrow \frac{udu}{u+1} = dx \quad \Rightarrow u - \log(u+1) = x$
$\Rightarrow \quad \log(x + y + 2) = y +$ **constant** $\Rightarrow x + y + 2 = Ce'$

Q.3 Answer:

$y^3 = -2x$

Solution:

$\left(xy^3 - x^2\right) dy - \left(xy + y^4\right) dx = 0$
$\Rightarrow \quad y^3(xdy - ydx) - x(xdy + ydx) = 0$
$\Rightarrow \quad x^2y^3 \frac{(xdy - ydx)}{x^2} - x(xdy + ydx) = 0$
$\Rightarrow \quad x^2y^3 d\left(\frac{y}{x}\right) - xd(xy) = 0$
Dividing by x^3y^2**, we get** $\frac{y}{x} d\left(\frac{y}{x}\right) - \frac{d(xy)}{x^2y^2} = 0$
Now, integrating $\frac{1}{2}\left(\frac{y}{x}\right)^2 + \frac{1}{xy} = c$
It passes through the point $(4, -2). \Rightarrow \quad \frac{1}{8} - \frac{1}{8} = c \Rightarrow c = 0 \quad \therefore \quad y^3 = -2x$

Q.4 Answer:

$y = \frac{x^2}{2} + x - \sin x$

Solution:

The differential equation is $\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2} = 1 + \sin \mathrm{x}$
Integrating we get $\frac{\mathrm{dy}}{\mathrm{dx}} = \mathrm{x} - \cos \mathrm{x} + \mathrm{c} \quad \ldots.$
When $x = 0$, $\frac{dy}{dx} = 0 \Rightarrow c = 1$
$\therefore$ **Equation (ii) is** $\frac{dy}{dx} = x - \cos x + 1$
Integrating again we get $y = \frac{x^2}{2} - \sin x + x + D$
When $\mathrm{x} = 0, \mathrm{y} = 0 \Rightarrow \mathrm{D} = 0$
$\therefore$ **The particular solution is** $\mathrm{y} = \frac{\mathrm{x}^2}{2} + \mathrm{x} - \sin \mathrm{x}$

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Q.5 Answer:

$\log\left((y+3)^2+(x+2)^2\right)+2\tan^{-1}\frac{y+3}{x+2}=C$

Solution:

The intersection of $y-x+1=0$ **and** $y+x+5=0$ **is** $(-2,-3)$**. Put** $x=X-2, y=Y-3$**. The given equation reduces to** $\frac{dY}{dX}=\frac{Y-X}{Y+X}$**. This is a homogeneous equation. Putting** $Y=vX$**, we get** $X\frac{dv}{dX}=-\frac{v^2+1}{v+1}$

$\Rightarrow \left(-\frac{v}{v^2+1}-\frac{1}{v^2+1}\right)dv=\frac{dX}{X}$

$\Rightarrow \quad -\frac{1}{2}\log\left(v^2+1\right)-\tan^{-1}v=\log|X|+$ **constant**

$\Rightarrow \log\left(Y^2+X^2\right)+2\tan^{-1}\frac{Y}{X}=$ **constant**

$\Rightarrow \quad \log\left((y+3)^2+(x+2)^2\right)+2\tan^{-1}\frac{y+3}{x+2}=\mathrm{C}$

Q.6 Answer:

$y=x\sin(\log x)$

Solution:

The given equation can be rewritten as $x^2\left[\frac{xdy-ydx}{x^2}\right]=x\sqrt{1-\frac{y^2}{x^2}}$

$\Rightarrow x^2\frac{d}{dx}\left(\frac{y}{x}\right)=x\sqrt{1-\frac{y^2}{x^2}}$

Put $\frac{y}{x}=z$**, we get** $x^2\frac{dz}{dx}=x\sqrt{1-z^2}\Rightarrow\frac{dz}{\sqrt{1-z^2}}=\frac{dx}{x}$

Integrating, we get $\sin^{-1}(z)=\log x+c\Rightarrow\sin^{-1}\left(\frac{y}{x}\right)=\log x+c$

Apply the boundary value $\mathrm{y}(1)=0\Rightarrow\sin^{-1}\left(\frac{0}{1}\right)=\log 1+\mathrm{c}\Rightarrow\mathrm{c}=0$

$\therefore \sin^{-1}\left(\frac{\mathrm{y}}{\mathrm{x}}\right)=\log\mathrm{x}+0\Rightarrow\mathrm{y}=\mathrm{x}\sin(\log\mathrm{x})$

Q.7 Answer:

$ke^{x^2/2}$

Solution:

Put $\mathrm{xy}=\mathrm{v}$

$\therefore \quad y+x\frac{dy}{dx}=\frac{dv}{dx}\Rightarrow\frac{dv}{dx}=x\frac{\phi(v)}{\phi'(v)}$

$\therefore \quad \frac{\phi'(v)}{\phi(v)}\mathrm{dv}=xdx$**. Integrating, we get**

$\log\phi(v)=\frac{x^2}{2}+\log k\Rightarrow\log\frac{\phi(v)}{k}=\frac{x^2}{2}$

or $\phi(v)=ke^{x^2/2}\Rightarrow\phi(xy)=ke^{x^2/2}$

Q.8 Answer:

$\log\left(\frac{y}{x}\right)=cx$

Solution:

The given equation can be written as $\frac{dy}{dx}=\frac{y}{x}\left(\ln\frac{y}{x}\right)+1$**. Substituting** $u=\frac{y}{x}$ **gives the differential equation** $\mathrm{x}\frac{\mathrm{du}}{\mathrm{dx}}=$ **ulnu whose solution is** $\ln\boldsymbol{u}=\boldsymbol{cx}$**.**

Q.9 Answer:

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$\phi(y/x) = kx$

Solution:

Let $\frac{y}{x} = v$

$\therefore \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$

Then, $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi(y/x)}{\phi'(y/x)}$ **reduces to**

$v + x\frac{dv}{dx} = v + \frac{\phi(v)}{\phi'(v)} \Rightarrow \frac{\phi'(v)}{\phi(v)}dv = \frac{dx}{x}$

On integrating, we get $\log \phi(v) = \log x + \log k$

$\Rightarrow \quad \phi(v) = xk \Rightarrow \phi\left(\frac{y}{x}\right) = kx$

Q.10 Answer:

$3/2$

Solution:

$\because y = u^n$

$\therefore \frac{dy}{dx} = nu^{n-1}\frac{du}{dx}$

On substituting the values of y **and** $\frac{dy}{dx}$ **in the given equation, then** $2x^4 \cdot u^n \cdot nu^{n-1}\frac{du}{dx} + u^{4n} = 4x^6$

$\Rightarrow \quad \frac{du}{dx} = \frac{4x^6 - u^{4n}}{2nx^4u^{2n-1}}$

Since, it is homogeneous. Then, the degree of $4x^6 - u^{4n}$ **and** $2nx^4u^{2n-1}$ **must be same.** $\therefore$

$4n = 6$ **and** $4 + 2n - 1 = 6$

Then, we get $n = \frac{3}{2}$

Q.11 Answer:

$\sin^{-1} y - \sin^{-1} x = c$

Solution:

$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}} \quad \therefore \frac{dy}{\sqrt{1-y^2}} = \frac{dx}{\sqrt{1-x^2}}$

$\Rightarrow \int \frac{dy}{\sqrt{1-y^2}} = \int \frac{dx}{\sqrt{1-x^2}} \Rightarrow \sin^{-1} y = \sin^{-1} x + c$

$\therefore \sin^{-1} y - \sin^{-1} x = c$

Q.12 Answer:

$\tan x$

Solution:

Given $\frac{dy}{dx} = 1 + y^2$

$\Rightarrow \frac{dy}{1+y^2} = dx \Rightarrow \tan^{-1} y = x + c$

$\Rightarrow y = \tan(x + c) \quad$ **Now** $y(0) = 0 \Rightarrow \tan c = 0$

$y(\pi) = 0 \Rightarrow \tan(\pi + c) = 0 \Rightarrow c = n\pi$

$\therefore y = \tan x$

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Q.13 Answer:

$\frac{c}{x^3}e^{-\frac{1}{x}}$

Solution:

The equation is $x\int_0^x y(t)dt = (x+1)\int_0^x ty(t)dt$ **.... (i)**

Differentiating both the sides with respect to x we get $xy(x) + \int_0^x y(t)dt = (x+1)xy(x) + \int_0^x ty(t)dt$

$\Rightarrow \int_0^x y(t)dt = x^2y(x) + \int_0^x ty(t)dt$

..... (ii)

Differentiating again with respect to x, we get $y(x) = 2xy(x) + x^2y'(x) + xy(x)$

$\Rightarrow x^2\frac{dy}{dx} = (1-3x)y$ **[writing** $y(x) = y$**]**

$\Rightarrow \frac{dy}{y} = \left(\frac{1-3x}{x^2}\right)dx = \left(\frac{1}{x^2} - \frac{3}{x}\right)dx$

Integrating we get, $\log y = -\frac{1}{x} - 3\log x + a$**, a is constant** $\Rightarrow \log y + 3\log x = a - \frac{1}{x}$

$\Rightarrow \log\left(yx^3\right) = a - \frac{1}{x} \Rightarrow yx^3 = e^{a-\frac{1}{x}} = c.e^{-\frac{1}{x}}$

where $c = e^a \quad \therefore y = \frac{c}{x^3}e^{-\frac{1}{x}}$

Q.14 Answer:

$n^2y + \cos nx = n^2(Cx + D)$

Solution:

The differential equation is $\frac{d^2y}{dx^2} = \cos nx$

Integrating

we get $\frac{dy}{dx} = \frac{\sin nx}{n} + C$

Integrating again $y = -\frac{\cos nx}{n^2} + Cx + D$

$\Rightarrow n^2y + \cos nx = n^2(Cx + D)$

Q.15 Answer:

$y = x\sin^{-1}a + c$

Solution:

$$\sin\left(\frac{dy}{dx}\right) - a = 0$$

$$\sin\left(\frac{dy}{dx}\right) = a \Rightarrow \frac{dy}{dx} = \sin^{-1}a; \quad dy = \sin^{-1}adx$$

Now integrating both sides, $\int dy = \int \sin^{-1}adx$

$$y = x\sin^{-1}a + c$$

Q.16 Answer:

$y + e^{-x} = c$

Solution:

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Consider the given differential equation $\ln\left(\frac{dy}{dx}\right)+x=0 \Rightarrow \ln\left(\frac{dy}{dx}\right)=-x$

$\Rightarrow \frac{dy}{dx}=e^{-x}$

on separating the variables, we get $dy=e^{-x}dx$

on integrating both sides, we get $\int dy=\int e^{-x}dx$

$\Rightarrow y=\frac{e^{-x}}{-1}+C=-e^{-x}+C \Rightarrow y+e^{-x}=C$

Q.17 Answer:

$1/3$

Solution:

$\frac{dy}{dx}\left(\frac{2+\sin x}{1+y}\right)=-\cos x, y(0)=1$

$\Rightarrow \quad \frac{dy}{(1+y)}=\frac{-\cos x}{2+\sin x}dx$

Integrating both sides $\Rightarrow \quad \ln(1+y)=-\ln(2+\sin x)+C$

Put $x=0$ **and** $y=1 \Rightarrow \ln(2)=-\ln 2+C \Rightarrow C=\ln 4$

Put $x=\frac{\pi}{2}$

$\ln(1+y)=-\ln 3+\ln 4=\ln\frac{4}{3} \Rightarrow y=\frac{1}{3}$

Q.18 Answer:

None of these

Solution:

$\frac{dt}{dx}=x^2-t \Rightarrow \frac{dt}{dx}+t=x^2$

$\Rightarrow t=\int x^2e^x dx+c \Rightarrow t=(x-1)^2e^x+c$

Q.19 Answer:

$y\sin y=x^2\log x+c$

Solution:

Given equation is

$(y\cos y+\sin y)dy=(2x\log x+x)dx$

$\Rightarrow y\sin y-\int 1\cdot\sin y dy+\int \sin y dy$

$=2\left[\log x\frac{x^2}{2}-\int\frac{1}{x}\cdot\frac{x^2}{2}dx\right]+\frac{x^2}{2}$

$\Rightarrow \mathrm{y}\sin \mathrm{y}=\mathrm{x}^2\log \mathrm{x}+\mathrm{c}$

Q.20 Answer:

$\sqrt{(x^2+y^2)}=a\sin\left\{\left(\tan^{-1}y/x\right)+\text{ constant }\right\}$

Solution:

Taking $x=r\cos\theta$ **and** $y=r\sin\theta$, **so that** $x^2+y^2=r^2$ **and** $\frac{y}{x}=\tan\theta$,

we have $xdx+ydy=rdr$

and $xdy-ydx=x^2\sec^2\theta d\theta=r^2d\theta$

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The given equation can be transformed into $\frac{rdr}{r^2 d\theta} = \sqrt{\left(\frac{a^2-r^2}{r^2}\right)}$

$\Rightarrow \quad \frac{dr}{d\theta} = \sqrt{(a^2 - r^2)} \Rightarrow \frac{dr}{\sqrt{(a^2-r^2)}} = d\theta$

Integrating both sides, then we get $\sin^{-1}\left(\frac{r}{a}\right) = \theta + c$

$\Rightarrow \quad \sin^{-1}\left(\frac{\sqrt{(x^2+y^2)}}{a}\right) = \tan^{-1}\left(\frac{y}{x}\right) + c$ **...(i)**

$\Rightarrow \quad \sqrt{(x^2+y^2)} = a \sin\left\{\tan^{-1}(y/x) + c\right\}$

$\Rightarrow \quad \sqrt{(x^2+y^2)} = a \sin\left\{\tan^{-1}(y/x) + \text{ constant }\right\}$

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